

Research Project Description

Explainable Machine Learning with Hybrid Feature Selection for Thyroid Disease Classification: A Case Study in Bangladesh

1. Project Overview

This project develops a leakage-safe and explainable machine-learning framework for five-class thyroid disease classification using locally collected clinical data from Bangladesh. The work targets automated support for classifying patients into euthyroid, clinical hyperthyroid, clinical hypothyroid, subclinical hyperthyroid, and subclinical hypothyroid categories. The study combines exploratory data analysis, rigorous preprocessing, hybrid feature selection, class-imbalance handling, ensemble learning, reliability assessment, and explainable AI to produce a clinically interpretable decision-support framework.

2. Clinical and Technical Motivation

- Thyroid disorders are common endocrine conditions and require accurate early diagnosis to reduce long-term health complications.
- Clinical thyroid diagnosis involves biochemical markers such as TSH, FT3, and FT4, but real-world cases can be complicated by symptom overlap and subclinical conditions.
- Many previous studies report high accuracy on public benchmark datasets, but often lack local clinical data, leakage-safe validation, multi-class diagnosis, calibration, and explainability.
- The proposed framework addresses these limitations by using patient-disjoint splitting, training-fold-only preprocessing, VT-MI feature selection, SMOTE within training folds, and SHAP/LIME explanation.

3. Research Objectives

- Build a machine-learning pipeline for five-class thyroid disease classification using Bangladesh clinical patient records.
- Reduce redundant predictors while preserving diagnostic performance through Variance Threshold and Mutual Information Gain (VT-MI) feature selection.
- Compare conventional, tree-based, boosting, bagging, voting, and stacked classifiers under the same leakage-safe protocol.
- Evaluate robustness using stratified cross-validation, independent hold-out testing, learning curves, bootstrap confidence intervals, calibration analysis, SHAP, and LIME.

4. Proposed Methodological Pipeline

Stage	Description
Data source	Clinical thyroid records collected through NINMAS, Dhaka, Bangladesh, with ethical approval and informed consent.
Target task	Supervised five-class classification: EU, HYPER, HYPO, SUB-HYPER, and SUB-HYPO.
Leakage control	Patient-disjoint 70:30 train-test split; the hold-out test set was isolated before preprocessing and model selection.
Preprocessing	KNN imputation for numerical missing values, mode imputation for categorical variables, one-hot encoding, and z-score scaling.
Feature selection	VT removes low-variance variables; MI selects the top 15 clinically informative predictors inside the training pipeline.
Class balancing	SMOTE applied only inside training folds with k-neighbors = 2 to avoid synthetic test leakage.
Modeling	DT, LR, KNN, SVM, NB, RF, AdaBoost, Bagging, XGBoost, CatBoost, Voting, and Stacked Ensemble.
Explainability	SHAP for global/local attribution and LIME for patient-specific explanation.

5. Key Results

Experiment	Best model(s)	Main performance outcome
Full-feature baseline	CatBoost and Bagging	Test accuracy 96.15%, macro-F1 94.90%, weighted-F1 96.09%, Cohen's kappa 93.29%, MCC 93.37%.
VT-MI feature selection	Bagging Ensemble	Preserved 96.15% test accuracy while reducing dimensionality to 15 selected features.
Cross-validation stability	Random Forest / CatBoost / Bagging	Strong validation performance; Random Forest showed high CV stability, while Bagging gave the strongest balanced hold-out result with VT-MI.
Robustness analysis	VT-MI Bagging	Bootstrap accuracy mean 0.9612 with 95% CI 0.923-0.990; macro-F1 mean 0.9459 with 95% CI 0.887-0.991.
Interpretability	SHAP and LIME	TSH and FT4 were consistently identified as dominant predictors; FT3 and symptom/history variables acted as secondary contributors.

6. Main Contributions

- Introduces a Bangladesh clinical-data-based five-class thyroid disease classification framework.
- Uses leakage-safe preprocessing, feature selection, oversampling, and model evaluation to reduce optimistic bias.
- Shows that VT-MI feature selection can simplify the feature space without reducing Bagging Ensemble performance.
- Adds clinical transparency by linking model predictions to physiologically meaningful biomarkers through SHAP and LIME.
- Provides reliability evidence through learning curves, bootstrap confidence intervals, calibration analysis, and SHAP-LIME agreement.

7. Limitations and Future Scope

- The dataset contains only 348 records and comes from a single clinical source, so external validity remains limited.
- Subclinical thyroid categories remain harder to separate because they share overlapping biochemical and symptomatic profiles.
- Future work should validate the framework on larger, multi-center cohorts and assess prospective clinical performance before deployment.

Source note: This compact description was prepared from the submitted manuscript titled "Explainable Machine Learning with Hybrid Feature Selection for Thyroid Disease Classification: A Case Study in Bangladesh."